INCH-POUND
MIL-DTL-24643/25E
22 August 2002
SUPERSEDING
MIL-C-24643/25D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LSPI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight with circuit integrity)

First -	Copper conductor, nickel coated, AWG 18, class B stranding.
Second -	Extruded silicone rubber insulation. Minimum average wall thickness 0.018 inch.
Third -	Glass braid, (one colored black and one colored white for each pair of conductors).
Fourth -	Transparent braid covering. Two conductors (one black and one white) cabled together to form a pair with lay not greater than 2.25 inches.
Fifth -	Braided shield of AWG No. 34 or 36 uncoated copper over each pair. Minimum coverage of 85 percent.
Sixth -	Shield insulation over each braided shield, consisting of polyethylene terephthalate film, per NEMA FI 4, 0.001 inch thick, applied helically with 50 percent overlap.
Seventh-	A glass braid over each shield insulation. (Standard identification code applied by method 4.)
Eighth -	The specified number of pairs (see table I) cabled together with a lay not greater than 14 inches. Fillers saturated with non-charring silicone base compound shall be used as necessary, to form a firm, round cable.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:03Z
Check the source to verify that this is the current version before use.

Construction (non-watertight with circuit integrity) (cont.)

Ninth - A binder tape applied helically with overlap.
Tenth - Silicone rubber jacket colored orange or red. (see table I for thickness).
Eleventh - Braided metal armor.

TABLE I. Details.

Military part no. M24643/25	Type and size	Number of pairs	Cable jacket thickness min avg (inch)	Overall diameter maximum (inch)	Cold bending mandrel diameter maximum (inches)
-01AN	LSPI-3	3	0.050	0.685	9.0
-02AN	LSPI-7	7	0.063	0.900	11.0
-03AN	LSPI-12	12	0.063	1.155	14.0

EXAMINATION AND TESTS:

Basic electrical: Requirements:

Conductor resistance - ohms/1000 feet at 25°C, maximum 7.5
Voltage withstand - volts, root mean square, minimum
Conductor to conductor 1200
Shield to shield 500
Insulation resistance - megohms -1000 feet, minimum...
Conductor to shield 500
Shield to shield 100
Conductor and shield continuity ..... No failure

Group A:

Visual and dimensional ..... No failure
The nickel coating on the conductors of the completed cable shall not be nicked, scraped or abraded so as to expose the copper conductor ..... No failure

Group B:

Cross-linked proof test (percent, maximum)
Insulation 50
Jacket (When tested at 200°C) 50
Gas flame - 1 hour at 120/70 volt root mean square ..... No failure
Cold bending, cable - (see table I for mandrel diameter) ..... No damage

Physicals (unaged)

Insulation
Tensile strength - lb/in^2, minimum 750
Elongation - percent, minimum 125
Jacket (cable)
Tensile strength - lb/in^2, minimum 800
Elongation - percent, minimum 250
Durometer (type A) hardness, minimum (ASTM D 2240) 60

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:03Z
Check the source to verify that this is the current version before use.

Group C:
Physicals - cable aging (260 ± 10°C)
Jacket
Tensile strength - lb/in^2, minimum 600
Elongation - percent, minimum 150
Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements:
Gas flame (3 hours) at 120/70 volts RMS	No failure
Aging and compatibility (cable) (125 ± 5°C)	No failure
For this test, omit all references to watertightness and fillers.	
Armor - conformance to material, construction and coverage	No failure
Specific gravity of extruded silicone insulation, maximum	1.55
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Smoke index, maximum	
Jacket	35
Fillers	45
Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5

UNIT ORDERING LENGTH:

All types and sizes 100 feet minimum.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-023)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:03Z
Check the source to verify that this is the current version before use.